

Lego Spike: — Bowling Fun

Today we will learn, build, test, and show our thinking.

Grade 4 learners: big steps, clear checks, and one thing at a time.

Today's Goal

- ▶ I can predict outcomes of the changes in energy that occur when objects collide.
- ▶ I can observe and describe the relationship between energy and force.
- ▶ I can engage effectively in a range of collaborative discussions.

Remember

We do not have to be perfect right away. We take one step, then the next step, then we check.

What You Need

- ▶ Your class materials
- ▶ The digital tool or build materials for today
- ▶ A partner voice and your thinking

Partner Norms

Look. Listen. Try. Talk. Help without taking over.

Step 1

Set up the robot and agree on partner roles.

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- ▶ One partner drives the device, one partner notices what to fix, then switch.
- ▶ Assign partner roles before materials come out so turn-taking is visible.

Pause. Check. Then move on.

Step 2

Build or code one move sequence at a time.

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- ▶ Test one change before adding another.
- ▶ Model one test-revise cycle with the actual robot or kit students will use.

Pause. Check. Then move on.

Step 3

Revise after each test and explain your choice.

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- ▶ Use We changed this because...
- ▶ Use a simple data language such as what changed, what happened, and what to try next.

Pause. Check. Then move on.

If You Get Stuck

1. Read the directions on your screen or slide one more time.
2. Check the example, the checklist, or your partner role.
3. Try one small change before asking for a full rescue.
4. Device-to-robot pairing and setup can split attention quickly.

Help Routine

Stop. Breathe. Point to where you are stuck. Ask a specific question.

Show Your Learning

- ▶ I stayed with the steps and used the help routine when I needed it.
- ▶ I made, coded, explained, or revised something that matches the lesson goal.
- ▶ I can point to one choice, fix, or idea I learned today.

Before You Leave

Save it, share it, explain it, or demonstrate it.

Teacher Voiceover Cues

- ▶ Today we are working on lego spike: — bowling fun. Watch first, then try it with me.
- ▶ If you feel stuck, stop at the checkpoint slide and use the help routine before you panic.
- ▶ Show what you learned by saving, sharing, explaining, or demonstrating one clear success move.

This slide can be removed before student use if you only want the visual deck.